



Shivansh Malik

Computer Science Undergraduate · Cyber Security & Applied AI
Building intelligent, security-minded systems end to end.

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PROFILE

Cyber Security-focused Computer Science undergraduate (CGPA 8.42/10) who designs and ships intelligent, security-minded systems end to end – from ML-driven intrusion detection to predictive healthcare, database-backed platforms, and applied Generative AI. Strong in data structures & algorithms, computer networks, operating systems, and database systems, with hands-on coursework spanning artificial intelligence, data analytics, and discrete mathematics. Curiosity spans probability, quantitative finance, and quantum computing. A national-level athlete and club lead who brings discipline, clear communication, and initiative to every team.

EDUCATION

YEAR	DEGREE / CERTIFICATE	INSTITUTE	CGPA/%
2024–Present	B.Tech, Computer Science & Engineering — Cyber Security	Manipal Institute of Technology, MAHE, Bengaluru	8.42/10
2023	CBSE (XII), Senior Secondary	Woodrow Senior Secondary School, Bareilly	83%
2021	CBSE (X), Secondary	Kendriya Vidyalaya, IFFCO — Aonla, Bareilly	94%
Semester I	Semester II	Semester III	Semester IV
8.85	8.41	7.48	8.95

Academic Distinctions: A+ in Design & Analysis of Algorithms, Introduction to Artificial Intelligence, Object-Oriented Software Development Lab, and Data Visualisation; A in Database Systems Lab, Operating Systems Lab, Digital Systems Lab, and Programming for Problem Solving Lab.

Relevant Coursework:

Data Structures & Lab	Design & Analysis of Algorithms	Operating Systems & Lab
Database Systems & Lab	Object-Oriented Programming & Lab	Digital Systems & Computer Organization
Data Communication & Computer Networks	Discrete Mathematical Structures	Computational Mathematics I & II
Probability & Optimization	Introduction to Artificial Intelligence	Data Analytics & Visualisation

Additional Coursework (First-Year Engineering Foundation):

Applied Physics for Engineers	Applied Chemistry for Engineers	Fundamentals of Electrical Engineering
Fundamentals of Electronics	Basic Mechanical Engineering Science	Engineering Mechanics & Smart Buildings
Workshop Practice	Computer-Aided Engineering Graphics	Communication Skills in English
Environmental Studies	Universal Human Values & Professional Ethics	Human Rights & the Constitution

TECHNICAL SKILLS

Languages	C, C++, Java, Python, JavaScript, HTML/CSS, SQL
Gen AI & LLM Engineering	Prompt engineering, RAG, embeddings, Groq/Gemini/Ollama APIs, ChromaDB, Streamlit, Pydantic
Machine Learning	scikit-learn, XGBoost, Pandas, NumPy, Random Forest, SMOTE, feature engineering, model evaluation
Data Analytics & Visualization	Exploratory data analysis, Matplotlib, Seaborn, statistical analysis
Databases	MySQL, Oracle SQL, MongoDB — schema design, transactions & concurrency control, query optimization
Computer Networks	OSI & TCP/IP models, IPv6 addressing & subnetting, routing fundamentals, socket programming
Systems & OS	Linux internals, POSIX threads & IPC, process synchronization, shell scripting
Cyber Security	Network traffic analysis (Wireshark), vulnerability & threat awareness, secure system design, ML-based intrusion detection
DSA & Competitive Programming	Working through a structured DSA sheet in C++ (arrays, trees, graphs, DP, greedy) with CLRS as reference; active on LeetCode and Codeforces, targeting 300+ problems solved and a 1400+ CF rating
Probability, Optimization & Quant Math	Probability theory, statistical optimization, curiosity in quantitative finance and algorithmic trading concepts
Core CS Fundamentals	DSA, OOP, DBMS, OS, Computer Networks, AI/ML, Discrete Mathematics, Digital Systems & Computer Organization
Tools & Platforms	Git/GitHub, JavaFX, Wireshark, VS Code, Linux/Unix CLI
Strengths	Leadership, communication, teamwork, public speaking, mentoring

EXPERIENCE

Generative AI Intern — Softpro School of AI

SSAI-101 · 45-Day / 6-Week AI Development Summer Training

Jun 2026 – Jul 2026 (Expected)

Remote

OBJECTIVE	Build practical, production-style Generative AI and LLM engineering skills across a structured 45-day curriculum, progressing from prompt engineering to retrieval-augmented generation and agentic workflows.
APPROACH	- Built and deployed multi-provider LLM applications on the Groq API (with Gemini/Ollama fallbacks), applying zero-shot, few-shot, and chain-of-thought prompting with Pydantic-validated structured outputs. - Implemented a semantic search and RAG pipeline from scratch using sentence-transformer embeddings and a Chroma vector database.
RESULT	Shipped the pipeline as Streamlit apps, including a PDF/URL AI content summarizer and a memory-persistent chatbot with session-state management.

Stack: Python, Groq/Gemini/Ollama APIs, Sentence-Transformers, ChromaDB, Streamlit, Pydantic

PROJECTS

Intrusion Detection System — ML on NSL-KDD

Apr 2026 – Present

Research Project · under Dr. Sanket Misra

OBJECTIVE	Build a modular, end-to-end ML pipeline flagging malicious network traffic on the NSL-KDD benchmark (41 features; binary & 5-class — Normal/DoS/Probe/R2L/U2R).
APPROACH	- Countered severe class imbalance with SMOTE; compared correlation, ANOVA-SelectKBest, and Random-Forest feature selection. - Benchmarked six classifiers to identify the strongest architecture for deployment. - Documented the preprocessing and evaluation methodology for reproducibility across binary and multi-class settings.
RESULT	Auto-selected and serialized the strongest model (~78% macro-F1) for deployment.

Stack: Python, scikit-learn, XGBoost, imbalanced-learn, Pandas

MedPredict — ML Symptom Checker & Hospital Locator

Dec 2025 – Apr 2026

OBJECTIVE	Build a Random-Forest engine that predicts likely conditions from reported symptoms, served through an interactive JavaFX desktop app.
APPROACH	- Integrated GPS geolocation to rank the nearest hospitals for the predicted condition. - Designed the JavaFX interface for real-time symptom input and result visualization.
RESULT	Delivered a fully working pipeline, model, and GUI — owned end to end individually.

Stack: Java, JavaFX, Random Forest, Geolocation

Concurrency-Safe Ticket Booking System

Dec 2025 – Apr 2026

DBMS Lab Project · Team of 6

OBJECTIVE	Design a normalized MySQL platform for high-demand event ticketing with a fully transactional booking workflow.
APPROACH	- Built conflict-resolution with transaction isolation and atomic seat reservation. - Modeled the schema across events, seats, and bookings with referential integrity constraints.
RESULT	Prevented seat double-booking under concurrent load across a 6-person team project.

Stack: MySQL, SQL, Transactions & Concurrency Control

POSITIONS OF RESPONSIBILITY

Educational Lead — Quantus, Quantum Computing Club

MIT Bengaluru

- Lead educational outreach for a flagship technical club — design and teach quantum-computing sessions for juniors, and help run events and workshops.
- Curate learning material bridging classical CS foundations with quantum computing concepts for a general undergraduate audience.
- Coordinate with faculty advisors to align club activities and session content with departmental academic goals.

Captain — Table Tennis Team

MIT Bengaluru

- Captain the institute squad — leading training, selection, and match strategy — to multiple tournament wins.
- Balance competitive-sport discipline with a rigorous academic schedule, translating athletic focus into consistent performance.
- Mentor junior players joining the institute team, focusing on technique, match temperament, and tournament readiness.

Member — Codex, Coding Club

MIT Bengaluru

- Sharpen data structures & algorithms with regular problem-solving; compete in hackathons and competitive-programming contests.
- Collaborate with peers on club-organized coding challenges and knowledge-sharing sessions.
- Contribute to peer code reviews and pair-programming sessions to reinforce good software practices.

CERTIFICATIONS

CERTIFICATION	ISSUER	YEAR
Linux on IBM LinuxONE	IBM SkillsBuild	2026
Integrating with Structured Databases	IBM SkillsBuild	2026
Data Fundamentals	IBM SkillsBuild	2025
Fundamentals of Java Programming	Board Infinity / Coursera	2025

ACHIEVEMENTS & EXTRA-CURRICULAR

Academic	- Class X Merit Scholarship from IFFCO for academic excellence. - NTSE Stage I — 150/200 (cleared cut-off); strong KVPY score. - A+ grades in Design & Analysis of Algorithms, Introduction to AI, OOP Software Development Lab, and Data Visualisation.
Sports	2× National-level Table Tennis player; captained MIT's team to multiple tournament titles.
Cultural	Professional Bharatanatyam dancer — represented Aonla twice, with a scholarship awarded.

INTERESTS

AI/ML · Cyber Security · Quantum Computing · Quantitative Finance · Competitive Programming · Chess · Reading History